Notes: Roberts and Tybout (AER, 1997)

The Decision to Export in Colombia: An Empirical Model of Entry with Sunk Costs

Contents

1 Big picture

- **Question.** Why do export flows respond unpredictably to exchange-rate movements and trade policy, and can this instability be explained by sunk costs of entering foreign markets?
- **Core mechanism.** If firms must incur a sunk cost to enter an export market, then current export participation depends on past export participation. Exporting experience becomes a state variable.
- **Empirical object.** The paper studies Colombian manufacturing plants from 1981 to 1989 and asks whether plant-level export status displays true state dependence after controlling for observed and unobserved plant heterogeneity.
- **Main test.** In a dynamic discrete-choice model, coefficients on lagged export participation identify sunk entry and reentry effects. If sunk costs are unimportant, past export participation should not affect current participation once plant characteristics and unobserved heterogeneity are controlled.
- **Main finding.** Prior exporting experience strongly raises the probability of exporting. The paper rejects the null that sunk costs are unimportant, and in some cases last year's export experience raises the probability of exporting by about 60 percentage points.
- **Experience depreciation.** The effect of export-market experience depreciates quickly. After two years out of the export market, reentry costs are statistically similar to the costs faced by a new exporter.
- **Policy implication.** Temporary devaluations or export-promotion policies may have persistent effects if they induce entry. But policies perceived as temporary may not induce entry because firms will not pay sunk costs unless expected future profits are high enough.
- **Economic takeaway.** Export supply is not a smooth static elasticity. It is a dynamic extensive-margin process shaped by entry costs, persistence, option value, and plant heterogeneity.

2 Incremental contribution of the paper

- **First plant-level test of sunk-cost hysteresis in exporting.** Earlier evidence on hysteresis relied mostly on aggregate trade flows, exchange rates, or sector-level export responses. Roberts and Tybout directly test the theory with plant-level export entry and exit decisions.
- **Separates true state dependence from spurious persistence.** A major contribution is distinguishing sunk-cost persistence from persistence caused by unobserved plant productivity, quality, managerial ability, or foreign contacts.
- **Dynamic discrete-choice framework for export participation.** The paper adapts dynamic binary-choice methods to international trade, modeling exporting as a state-dependent decision with lagged participation, permanent plant effects, and transitory shocks.
- **Initial-conditions correction.** Because lagged export status is endogenous to unobserved plant effects, the paper explicitly handles the initial-conditions problem using a presample approximation.
- **Quantifies experience depreciation.** The paper does not only show that experience matters; it estimates how quickly export-market experience loses value after exit.
- **Micro-foundation for unstable export supply elasticities.** The findings explain why aggregate export supply functions can be unstable across countries and periods: the response depends on how many plants are near entry thresholds and how many have recent export experience.
- **Policy contribution.** The paper shifts export-promotion policy from a pure price-subsidy view to an entry-margin view. Policies that help firms enter and learn export markets can have different effects than policies that raise the volume of existing exporters.
- **Methodological legacy.** The paper became a template for later empirical work on export sunk costs, export persistence, extensive-margin trade responses, and firm dynamics in international markets.

3 Data and measurement

3.1 Plant-level panel

- The paper uses annual plant-level data from the Colombian manufacturing census for 1981–1989.
- The census covers plants with 10 or more employees and reports location, industry, age, ownership, capital stock, investment, labor and materials expenditure, domestic sales, and export sales.
- The authors match plant observations over time to create a panel.
- The core dependent variable equals one if plant i exports in year t and zero otherwise.

Interpretation: The data allow the authors to observe actual entry and exit into export markets rather than inferring entry from aggregate trade flows.

3.2 Industry and estimation sample

- Table 1 covers 19 three-digit ISIC manufacturing industries that account for most Colombian manufacturing exports.
- The econometric model is estimated using four major exporting industries: food, textiles, paper, and chemicals.
- The final estimation panel contains 650 plants observed in every sample year, yielding 5,850 plant-year observations.
- Years 1981–1983 serve as presample years for initial conditions; 1984–1989 are used for estimation.

Interpretation: The estimation sample is not intended to represent all plants. It is chosen to follow established producers through export entry and exit decisions.

3.3 Macroeconomic and policy variables

- Table 1 documents real exchange rate movements, export quantities, export subsidy rates, number of exporting plants, and proportion of plants exporting.
- A higher real effective exchange-rate index corresponds to a depreciation of the Colombian peso.
- The export environment in the early 1980s was unfavorable: the peso appreciated through 1982, and import protection biased incentives toward domestic markets.
- After 1984, depreciation improved the export environment, but participation responses were not proportional.

Interpretation: These macro facts motivate the hysteresis question: why did large changes in the exchange rate produce modest or delayed changes in the number of exporters?

3.4 Export history variables

- $Y_{i,t-1}$ indicates whether the plant exported last year.
- $\widetilde{Y}_{i,t-2}$ and $\widetilde{Y}_{i,t-3}$ capture whether the plant last exported two or three years earlier.
- The coefficients on these variables measure the effect of prior export-market experience on current export participation.
- Declining coefficients across lags indicate depreciation of export-market experience.

Interpretation: Export history variables are the empirical bridge from sunk-cost theory to data. They test whether exporting experience changes current profitability after conditioning on other plant traits.

4 Models and empirical specifications

4.1 Theoretical entry condition

Let $\pi_i(P_t, s_{it})$ be the difference in expected gross profits between exporting and not exporting. A plant that has never exported faces an entry cost F_i^0. A plant that last exported j years ago faces reentry cost F_i^j. A plant that exported last year can continue without paying the new-entry cost.

The participation decision compares expected dynamic values:

$$Y_{it} = 1 \quad \text{if export value exceeds nonexport value.}$$

Interpretation: The decision is dynamic because exporting today changes future participation costs and therefore the continuation value.

4.2 Reduced-form expected profit equation

The paper approximates net expected export profitability with observed characteristics and unobserved shocks:

$$\pi_{it}^* - F_i^0 = \theta_t + \beta Z_{it} + \varepsilon_{it}.$$

- θ_t captures year or macroeconomic conditions.
- Z_{it} includes wages, export prices, capital stock, age, ownership, industry, and region.
- ε_{it} captures unobserved profitability shocks.

Interpretation: This equation avoids imposing a full structural profit function while still controlling for observables that affect export profitability.

4.3 Dynamic probit estimating equation

The central estimating equation is a dynamic binary-choice model:

$$Y_{it} = 1\left[\theta_t + \beta Z_{it} + \gamma_0 Y_{i,t-1} + \sum_{j=2}^{J} \gamma_j \widetilde{Y}_{i,t-j} + \varepsilon_{it} > 0\right].$$

- γ_0 captures the effect of having exported last year.
- γ_j captures the lingering value of export experience from j years ago.
- If sunk costs are irrelevant, the lagged export variables should be jointly zero.

Interpretation: The model translates sunk costs into estimable coefficients on lagged export history.

4.4 Unobserved heterogeneity and serial correlation

The unobserved component is decomposed into a permanent plant effect and a transitory shock:

$$\varepsilon_{it} = \alpha_i + w_{it}, \qquad w_{it} = \rho w_{i,t-1} + \eta_{it}.$$

- α_i captures persistent plant traits such as managerial quality, export contacts, product quality, or efficiency.
- w_{it} captures transitory shocks to export profitability.
- Controlling for α_i is essential because otherwise persistent heterogeneity could be mistaken for sunk costs.

Interpretation: This is the paper's main identification device: true state dependence must be separated from unobserved plant-level persistence.

4.5 Initial-conditions problem

Because lagged export status in the first estimation years is correlated with unobserved plant effects, the paper uses a presample approximation for export participation in 1981–1983.

$$Y_{it}^p = 1[\lambda Z_{it}^p + \varepsilon_{it}^p > 0].$$

Interpretation: This correction prevents the model from treating presample export status as exogenous when it is partly determined by the same unobserved plant traits that affect later exporting.

4.6 Estimation methods

- The paper estimates the model by method of simulated moments (MSM) and maximum likelihood (ML).
- Different specifications allow three export lags, one export lag, permanent plant effects, and serially correlated transitory shocks.
- The main preferred specification is Model 2, which includes three years of export history and permanent plant heterogeneity.

Interpretation: Similar conclusions across MSM and ML strengthen the evidence that lagged exporting matters beyond unobserved heterogeneity.

5 Identification and empirical design

5.1 Core identification challenge

Export status is persistent. But persistence can be caused by two very different forces:

1. **True state dependence:** exporting experience lowers future costs because entry costs are sunk.
2. **Spurious state dependence:** some plants are always more likely to export because they are more productive, older, better managed, or located closer to markets.

Interpretation: The paper's central contribution is to distinguish these two explanations.

5.2 How the model identifies sunk costs

- Observed plant characteristics control for measured differences in export profitability.
- Permanent random effects control for unobserved plant differences.
- Lagged export-status coefficients capture remaining state dependence.
- A joint test that lagged export-history coefficients equal zero is a test for the absence of sunk-cost effects.

Interpretation: If lagged exporting remains important after controlling for plant heterogeneity, this is evidence of sunk entry or reentry costs.

5.3 Why multiple lags matter

- A one-lag model can show persistence but cannot measure how quickly experience depreciates.
- The three-year lag model distinguishes last-year exporters from plants that last exported two or three years ago.
- The results show that the most recent export experience matters strongly, while experience two or three years old has much weaker effects.

Interpretation: The depreciation pattern helps interpret sunk costs as market-specific knowledge, contacts, or product adaptation whose value decays after exit.

5.4 Model fit and validation

- Table 4 compares predicted and observed broad export trajectories.
- Appendix Table A1 compares predicted and actual frequencies across all 64 possible export histories for 1984–1989.
- The model matches broad transitions well, though very detailed trajectory tests are more demanding.

Interpretation: The specification is not only judged by coefficient significance; it is also evaluated by whether it reproduces observed export histories.

5.5 Limitations

- The model is reduced-form in export profitability and does not estimate a full structural profit function.
- It studies established plants in four major exporting industries, not all Colombian manufacturers.
- Sunk costs are inferred from state dependence rather than directly observed.
- Export intensity is not modeled; the dependent variable is participation.

Interpretation: These limitations are important, but the paper's design is powerful for identifying whether entry history affects export participation.

6 Section-by-section study guide

- **Introduction.** Frames export supply instability and sunk-cost hysteresis; explains why aggregate evidence is inadequate.
- **Section I: Theory.** Derives the dynamic export participation condition with entry costs, reentry costs, exit costs, and experience depreciation.
- **Section II: Data and Colombian context.** Describes plant-level census data, macroeconomic policy regime, and transition rates.
- **Section III: Econometric model.** Develops the dynamic probit, unobserved heterogeneity structure, initial-conditions correction, and estimation strategy.

- **Section IV: Results.** Reports sunk-cost coefficients, plant-characteristic effects, model fit, and predicted export probabilities.
- **Conclusion.** Draws policy implications for export-promotion strategies and export-market entry.

7 Tables and appendix table

7.1 Tables 1 and 2: Macro export environment and export-status transitions

Read:

- Table 1 shows exchange rates, exports, subsidies, number of exporters, and exporter shares from 1981 to 1989.
- The real exchange rate depreciates sharply after 1984, and exports rise, but the proportion of plants exporting responds only gradually.
- Table 2 shows strong persistence in export status. Nonexporters mostly remain nonexporters, and exporters mostly remain exporters.
- Average nonexporter-to-exporter transition rates are around 3 percent, while exporter-to-exporter persistence is around 87 percent for the 19-industry sample.

Detailed interpretation:

- Table 1 motivates the empirical puzzle: large exchange-rate and policy changes do not mechanically translate into equally large exporter-entry responses.
- Table 2 supplies preliminary evidence of hysteresis. Export participation is persistent, but the table alone cannot tell whether this is true sunk-cost state dependence or unobserved heterogeneity.
- Together, the tables justify the dynamic plant-level model: macro conditions matter, but exporting history and plant-level transitions are central.

Economic message: Persistence is the phenomenon to explain. The rest of the paper asks whether this persistence reflects sunk costs rather than simply stable differences across plants.

TABLE 1—COLOMBIAN MANUFACTURED EXPORTS 1981–1989 (19 THREE-DIGIT ISIC INDUSTRIES)

Variable	1981	1982	1983	1984	1985	1986	1987	1988	1989
Real effective exchange rate index (1975 = 100)[a]	84.0	79.5	80.5	89.8	102.2	113.6	113.7	112.3	115.3
Quantity of exports (1986 = 100)[b]	58.5	64.1	63.8	59.1	66.4	100.0	88.6	103.1	127.2
Export subsidy rate	0.055	0.055	0.066	0.099	0.092	0.047	0.047	0.042	0.044
Number of exporting plants	667	676	615	585	653	705	707	735	816
Proportion of plants that export	0.129	0.128	0.113	0.107	0.117	0.112	0.119	0.124	0.135

[a]Source: José Ocampo and Leonardo Villar (1995). An increase in this variable corresponds to a devaluation of the Colombian peso.
[b]Source: Roberts et al. (1995). Figures describe export-oriented industries only.

(a) Table 1 - Colombian Manufactured Exports 1981-1989

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TABLE 2—PLANT TRANSITION RATES IN THE EXPORT MARKET 1982–1989

Year-t status	Year-(t + 1) status	1982–1983	1983–1984	1984–1985	1985–1986	1986–1987	1987–1988	1988–1989	Average, 1982–1989
A. *Nineteen three-digit manufacturing industries:*									
No exports	No exports	0.974	0.971	0.957	0.963	0.973	0.972	0.958	0.967
	Exports	0.026	0.029	0.043	0.037	0.026	0.028	0.042	0.033
Exports	No exports	0.168	0.135	0.131	0.108	0.158	0.086	0.107	0.128
	Exports	0.832	0.865	0.869	0.892	0.842	0.914	0.893	0.872
B. *Four major exporting industries:*									
No exports	No exports	0.971	0.969	0.972	0.960	0.983	0.972	0.985	0.973
	Exports	0.029	0.031	0.028	0.040	0.017	0.028	0.015	0.027
Exports	No exports	0.108	0.101	0.152	0.124	0.149	0.085	0.054	0.110
	Exports	0.892	0.899	0.848	0.876	0.851	0.915	0.946	0.890

ket, the proportion of manufacturing plants $\pi^*_{it} = \pi_i(\mathbf{p}_t, \mathbf{s}_{it})$

(b) Table 2 - Plant Transition Rates in the Export Market 1982-1989

7.2 Table 3: Dynamic Probit Model of Export Participation

Read:

- Table 3 reports MSM and ML estimates of dynamic probit models.
- The coefficient on last year's export status is positive and statistically significant across specifications.
- Two- and three-year lag effects are smaller, showing depreciation of export experience.
- Plant size, age, corporate ownership, industry dummies, and region are important predictors of exporting.
- The variance of permanent plant effects is large, indicating important unobserved heterogeneity.

Detailed interpretation:

- This is the core empirical table. The significant coefficient on $Y_{i,t-1}$ is the direct evidence that export-market experience affects current participation.
- The model controls for permanent plant effects, so the lagged-export coefficient is not simply a proxy for plant quality.
- The three-lag model shows that experience depreciates quickly. Exporting last year is much more valuable than having exported two or three years earlier.
- The permanent heterogeneity estimates also matter: sunk costs are not the only source of persistence. Plant traits and experience both shape participation.

Economic message: The evidence supports sunk-cost hysteresis while also showing that heterogeneous plants respond differently to the same macro conditions.

TABLE 3—DYNAMIC PROBIT MODEL OF EXPORT PARTICIPATION
(STANDARD ERRORS IN PARENTHESES)

	Model 1	Model 2		Model 3	
Explanatory variable	(i) MSM	(ii) MSM	(iii) ML	(iv) MSM	(v) ML
Intercept	−7.105*	−7.058*	−7.039*	−6.856*	−7.033*
	(1.222)	(1.215)	(1.021)	(1.149)	(0.988)
Y_{t-1}	1.036*	0.971	1.140*	0.702*	0.885*
	(0.326)	(0.261)	(0.211)	(−0.154)	(0.135)
$\tilde{Y}_{t-2}$	0.326	0.331	0.401*	—	—
	(0.190)	(0.181)	(−0.145)		
$\tilde{Y}_{t-3}$	0.069	0.068	0.130	—	—
	(0.182)	(0.176)	(−0.164)		
1985 dummy	−0.156	−0.160	−0.168	−0.140	−0.154
	(0.133)	(0.109)	(0.106)	(0.100)	(0.097)
1986 dummy	−0.013	−0.022	−0.026	−0.017	−0.021
	(0.111)	(0.106)	(0.108)	(0.093)	(0.098)
1987 dummy	−0.309*	−0.312*	−0.340*	−0.286*	−0.318*
	(0.109)	(0.107)	(0.112)	(0.098)	(0.103)
1988 dummy	−0.148	−0.161	−0.187	−0.155	−0.178
	(0.119)	(0.115)	(0.114)	(0.102)	(0.104)
1989 dummy	−0.313*	−0.322*	−0.355*	−0.305*	−0.343*
	(0.111)	(0.110)	(0.118)	(0.098)	(0.109)
$\ln(\text{Wage}_{t-1})$	0.142	0.136	0.174	0.115	0.173
	(0.127)	(0.126)	(0.107)	(0.116)	(0.101)
$\ln(\text{Export price}_{t-1})$	−0.029	−0.027	−0.065	−0.025	−0.059
	(0.055)	(0.055)	(0.046)	(0.052)	(0.047)
$\ln(K_{t-1})$	0.207*	0.211*	0.222*	0.207*	0.221*
	(0.032)	(0.032)	(0.031)	(0.031)	(0.033)
$\ln(\text{Age}_{t-1})$	0.471*	0.471*	0.349*	0.506*	0.396*
	(0.126)	(0.126)	(0.096)	(0.123)	(0.089)
Corporation	0.383*	0.386*	0.271*	0.450*	0.308*
	(0.156)	(0.152)	(0.115)	(0.148)	(0.111)
Textiles industry dummy	0.817*	0.815*	0.681*	0.839*	0.698*
	(0.159)	(0.159)	(0.135)	(0.158)	(0.158)
Paper industry dummy	0.310	0.305	0.165	0.319*	0.163
	(0.184)	(0.183)	(0.147)	(0.191)	(0.145)
Chemical industry dummy	0.762*	0.760*	0.640*	0.811*	0.689*
	(0.203)	(0.199)	(0.134)	(0.200)	(0.130)
Cali/Medellín	0.112	0.119	−0.017	0.119	−0.052
	(0.131)	(0.133)	(0.117)	(0.135)	(0.129)
Other region	0.479*	0.487*	0.453*	0.504*	0.471*
	(0.134)	(0.135)	(0.104)	(0.137)	(0.098)
$\text{Var}(\alpha)$	0.668*	0.687*	0.620*	0.764*	0.688*
	(0.119)	(0.096)	(0.078)	(0.061)	(0.051)
$\text{Corr}(\alpha,\alpha^{p})$	0.898*	0.894*	0.899*	0.906*	0.901*
	(0.039)	(0.038)	(0.017)	(0.032)	(0.017)
ρ	−0.019	—	—	—	—
	(0.028)				
Log likelihood:	−854.8[a]	−854.1[a]	−837.7	−857.8[a]	−842.700

* Statistically significant at the 5-percent level.
[a] Simulated with 1,000 draws of the errors.

Table 3 - Dynamic Probit Model of Export Participation

7.3 Tables 4 and 5: Model fit and predicted exporting probabilities

Read:

- Table 4 compares observed versus predicted broad export trajectories.

- The model predicts 0.737 always-nonexporters versus 0.763 observed, and 0.116 always-exporters versus 0.109 observed.
- Table 5 computes predicted probabilities for plants with different observables, export histories, and unobserved plant effects.
- For plants with favorable observable and unobservable characteristics, last-year exporting dramatically increases current export probability.

Detailed interpretation:

- Table 4 is a goodness-of-fit test. It shows the dynamic model can reproduce major export-participation paths well.
- Table 5 translates coefficients into economically interpretable probabilities. It shows that export history matters most for plants close to the participation margin.
- The table also clarifies heterogeneity: for plants with very low expected profitability, export experience alone is not enough to induce participation; for high-profitability plants, export experience can be decisive.

Economic message: Sunk costs create threshold behavior. Policy shocks matter most when they move plants across entry thresholds and when recent export experience still has value.

TABLE 4—OBSERVED VERSUS PREDICTED FREQUENCIES OF Y_{IT} TRAJECTORIES
(BASED ON ML ESTIMATES OF MODEL 2 IN TABLE 3)

Trajectory type	Observed frequencies	Predicted frequencies
Always a nonexporter	0.763	0.737
Begin as a nonexporter, switch once	0.045	0.044
Begin as a nonexporter, switch at least twice	0.029	0.033
Always an exporter	0.109	0.116
Begin as an exporter, switch once	0.022	0.037
Begin as an exporter, switch at least twice	0.032	0.034

(a) Table 4 - Observed versus Predicted Frequencies

TABLE 5—PREDICTED PROBABILITY OF EXPORTING
(BASED ON ML ESTIMATES OF MODEL 2 IN TABLE 3)

Plant effect	25th percentile of $\boldsymbol{\beta Z}_{it}$				50th percentile of $\boldsymbol{\beta Z}_{it}$				75th percentile of $\boldsymbol{\beta Z}_{it}$			
$(\alpha)^{a}$	$(Y_{t-1}, \tilde{Y}_{t-2}, \tilde{Y}_{t-3})=$ (0,0,0)	(0,0,1)	(0,1,0)	(1,0,0)	(0,0,0)	(0,0,1)	(0,1,0)	(1,0,0)	(0,0,0)	(0,0,1)	(0,1,0)	(1,0,0)
−2	0.000	0.000	0.000	0.003	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
−1	0.000	0.000	0.000	0.001	0.000	0.000	0.000	0.010	0.000	0.001	0.004	0.073
0	0.000	0.000	0.001	0.037	0.002	0.004	0.012	0.146	0.022	0.035	0.085	0.431
1	0.009	0.016	0.044	0.306	0.052	0.079	0.165	0.588	0.228	0.296	0.462	0.865
2	0.141	0.193	0.335	0.780	0.363	0.445	0.618	0.933	0.702	0.771	0.881	0.991

Note: Each table entry is the predicted probability of exporting for a given combination of export history (Y

(b) Table 5 - Predicted Probability of Exporting

7.4 Appendix Table A1: Predicted and Actual Frequencies

Read:

- Appendix Table A1 reports predicted and actual frequencies for the 64 possible export-status trajectories from 1984 to 1989.
- The largest category is plants that never export in any year: predicted 0.737 and actual 0.763.
- Some rare trajectories are difficult to match exactly, which is expected because many possible patterns are almost never observed.
- The appendix provides a detailed diagnostic beyond the broad trajectory categories in Table 4.

Detailed interpretation:

- Table A1 is useful because it checks whether the model reproduces the full sequence distribution, not only single-year transitions or broad trajectory groups.
- The fit is strongest for common histories, especially never-exporting and always-exporting paths.
- The detailed chi-square test is more demanding, partly because many cells are sparse. This does not overturn the central evidence in Table 3 but reminds the reader that dynamic discrete-choice models can fit common patterns better than rare transitions.

Economic message: The model is a useful approximation to export dynamics, especially for economically important participation histories.

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TABLE A1—PREDICTED AND ACTUAL FREQUENCIES OF Y_{it} TRAJECTORIES

		Export status[a]					
Predicted frequency	Actual frequency	1984	1985	1986	1987	1988	1989
0.116	0.109	1	1	1	1	1	1
0.007	0.003	1	1	1	1	1	0
0.003	0.002	1	1	1	1	0	1
0.004	0.003	1	1	1	1	0	0
0.007	0.005	1	1	1	0	1	1
0.001	0	1	1	1	0	1	0
0.002	0	1	1	1	0	0	1
0.007	0.005	1	1	1	0	0	0
0.002	0.003	1	1	0	1	1	1
0.000	0	1	1	0	1	1	0
0.000	0	1	1	0	1	0	1
0.000	0.002	1	1	0	1	0	0
0.001	0.003	1	1	0	0	1	1
0.000	0	1	1	0	0	1	0
0.001	0.002	1	1	0	0	0	1
0.006	0.002	1	1	0	0	0	0
0.005	0.006	1	0	1	1	1	1
0.001	0	1	0	1	1	1	0
0.000	0.002	1	0	1	1	0	1
0.001	0	1	0	1	1	0	0
0.001	0	1	0	1	0	1	1
0.000	0	1	0	1	0	1	0
0.000	0	1	0	1	0	0	1
0.001	0.006	1	0	1	0	0	0
0.001	0	1	0	0	1	1	1
0.000	0	1	0	0	1	1	0
0.000	0	1	0	0	1	0	1
0.000	0	1	0	0	1	0	0
0.002	0.002	1	0	0	0	1	1
0.000	0.002	1	0	0	0	1	0
0.001	0	1	0	0	0	0	1
0.012	0.009	1	0	0	0	0	0
0.008	0.014	0	1	1	1	1	1
0.002	0	0	1	1	1	1	0
0.000	0	0	1	1	1	0	1
0.001	0	0	1	1	1	0	0
0.001	0	0	1	1	0	1	1
0.000	0	0	1	1	0	1	0
0.001	0	0	1	1	0	0	1
0.002	0.002	0	1	1	0	0	0
0.000	0	0	1	0	1	1	1
0.000	0	0	1	0	1	1	0
0.000	0	0	1	0	1	0	1
0.000	0	0	1	0	1	0	0
0.000	0.002	0	1	0	0	1	1
0.000	0.002	0	1	0	0	1	0
0.000	0	0	1	0	0	0	1
0.002	0.003	0	1	0	0	0	0
0.009	0.006	0	0	1	1	1	1
0.002	0	0	0	1	1	1	0
0.001	0	0	0	1	1	0	1
0.002	0.002	0	0	1	1	0	0
0.001	0	0	0	1	0	1	1
0.001	0	0	0	1	0	1	0
0.001	0	0	0	1	0	0	1
0.006	0.012	0	0	1	0	0	0
0.005	0.008	0	0	0	1	1	1
0.002	0	0	0	0	1	1	0
0.000	0	0	0	0	1	0	1
0.002	0.003	0	0	0	1	0	0
0.011	0.011	0	0	0	0	1	1
0.005	0.005	0	0	0	0	1	0
0.011	0.006	0	0	0	0	0	1
0.737	0.763	0	0	0	0	0	0

[a] Export-market participation is denoted by 1's; nonparticipation is denoted by 0's.

Appendix Table A1 - Predicted and Actual Frequencies

8 Typed theoretical result blocks

8.1 Result block 1: Sunk entry and reentry costs

- A plant that never exported must pay a first-entry cost.
- A plant that exported in the past but exited may face a reentry cost.
- Reentry costs can depend on how long the plant has been absent from export markets.
- A continuing exporter does not need to recreate the same market access from scratch.

Interpretation: This is the core theory: past exporting affects current choices because it changes current and future costs.

8.2 Result block 2: True state dependence versus heterogeneity

- True state dependence means exporting last year causally changes today's export payoff.
- Heterogeneity means some plants are persistently more likely to export regardless of past participation.
- The empirical model includes permanent plant effects to separate these channels.

Interpretation: Without heterogeneity controls, a regression would overstate sunk costs by attributing high-quality plants' repeated exporting to export experience.

8.3 Result block 3: Policy nonlinearity

- Devaluations and subsidies will not affect all plants equally.
- Plants far below the export threshold do not enter, even after moderate shocks.
- Plants already close to the threshold can enter and then remain exporters after the shock disappears.

Interpretation: The aggregate export response depends on the distribution of plants around participation thresholds.

9 Detailed result map

- **Macro environment:** exchange-rate movements and export subsidies create time-varying incentives, but participation responses are slow and asymmetric.
- **Transition evidence:** export status is highly persistent, motivating a dynamic model.
- **Dynamic probit evidence:** lagged export status remains significant after controlling for plant characteristics and unobserved plant effects.
- **Experience depreciation:** recent export experience matters most; effects after two or three years are much smaller.
- **Heterogeneity:** plant size, age, corporate status, industry, region, and unobserved plant effects strongly affect exporting.
- **Model fit:** the model reproduces broad export trajectories and common detailed histories.
- **Policy implication:** export-promotion policies should distinguish helping current exporters expand from inducing new entry.

10 Short summary

- The paper tests whether sunk entry costs explain persistence and hysteresis in export participation.
- It uses Colombian manufacturing plant-level panel data from 1981 to 1989.
- It estimates a dynamic probit model of whether a plant exports in a given year.
- Lagged export participation identifies state dependence after controlling for observable and unobservable plant characteristics.
- Prior export experience significantly raises the probability of exporting.
- Export experience depreciates quickly after exit.
- Plant heterogeneity also matters: size, age, ownership, industry, region, and permanent plant effects all affect exporting.
- The findings support sunk-cost hysteresis models and explain why export supply responses are unstable across countries and time.

11 Conclusion

11.1 Core conclusion

Roberts and Tybout show that Colombian plants' decisions to export are strongly shaped by sunk entry costs and prior export-market experience. Export history matters even after controlling for observed plant characteristics and unobserved permanent plant heterogeneity.

11.2 Economic intuition

Entering export markets requires firms to build knowledge, relationships, product adaptations, and market access. Once these costs are paid, a firm is more likely to continue exporting. But if the firm exits, the value of this experience depreciates, so reentry eventually resembles first entry.

11.3 Incremental contribution revisited

The paper's incremental contribution is to bring the sunk-cost hysteresis theory to plant-level data using a dynamic discrete-choice model. It directly tests the theory's key micro implication: current export participation depends on prior export participation because entry experience changes costs.

11.4 Implications for trade policy

Export-promotion policies should not only subsidize volumes of existing exporters. If sunk costs are important, policies that help firms enter foreign markets, learn demand conditions, build contacts, and maintain export experience may have more persistent effects.

11.5 Limitations

The paper models the participation margin rather than export volumes or destinations. Sunk costs are inferred from dynamic behavior, not directly observed. The estimation sample focuses on established plants in four major export-oriented industries. These choices make the design sharp for entry costs but limit generality.

11.6 Takeaway

Export supply is a dynamic extensive-margin process. Temporary shocks can have persistent effects if they induce entry, but favorable conditions may have little effect if firms doubt their permanence or if they are far below the export threshold.